

Class #0 - How to prove a statement

1. Direct Proof:

- (a) Example: Prove: Let m and n be integers. If both m and n are odd, then mn is odd.
- (b) [If P then Q] or [P implies Q] or $[P \Rightarrow Q]$ (called "conditional statement")
P: hypothesis, Q: conclusion
- (c) Proof: Let _____. (State the Hypothesis)
To show: _____. (State the Conclusion: What would show this conclusion?
Have to know the definition of an _____.

2. Example: Prove: For every integer n , the integer $n^2 + n$ is even.
(Note: for "every" integer! What is needed?)

3. Indirect Proof: by Contradiction

The characteristic of Dr. Bean is that if she arrives in the morning at the Math department, she takes her mug from the department office and take a cup of coffee, and she is never without her coffee until she goes home in the evening.

- (a) Example: *Prove: Dr. Bean has not arrived yet.*

Assume the contrary and show this would arrive to a contradiction to either its characteristics or definitions.

- (b) Example: *Prove: There is no smallest positive rational number.*

Proof: Suppose _____.

What is the definition that characterizes _____?

- (c) Example: *Prove: Let m and n be integers. Prove that if mn is odd, then both m and n are odd.*

”without loss of generality” WLOG

4. Indirect proof: Proving the contrapositive

- (a) $P \Rightarrow Q$ is logically equivalent to _____.

Example: *If Phil parties until 3am Sunday night, then Phil looks tired Monday* is logically equivalent to

_____.

- (b) Example: Let x be a positive real number. Prove that if x is irrational, then is $\sqrt{x}$ irrational.

Proof: Will show _____ (**)

We can show (**) by?

- (c) Example: Suppose the points (x_1, y_1) and (x_2, y_2) each satisfy the equation $y = mx + b$. Prove that if $y_1 \neq y_2$, then $x_1 \neq x_2$.

5. To prove a statement that is true for each positive integer n , the proof technique, **Mathematical Induction**, is devised:

Use properties of Natural numbers:

A list that starts with 1 satisfying the following property (*)

((*): given any number in the list (call it n), its successor, $n + 1$, is also in the list),
includes all natural numbers.

6. Example: Show $\sum_{i=1}^n (2i - 1) = n^2$ for every positive integer n .